University of Chittagong
Department of Computer Science and Engineering
8th Semester B.Sc. Engg. Examination - 2021
Course Code: CSE 815
Course Title: Machine Learning
Total Marks: 52.5 Marks Time: 4:00 Hours

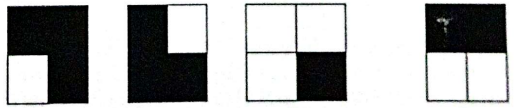
[Answer the six of the following questions Three from each Section A and Section B; Figures in the right-hand margin indicate full marks.]

Section A

1. (a) What is "Machine Learning"? What is dimension reduction in Machine Learning? Give examples. **2.75**
   (b) Describe different types of Machine learning problems with necessary examples. **3**
   (c) Describe different types of Machine learning methods with necessary examples. Make a comparison among different machine learning methods. **3**

2. (a) Define (i) False Acceptance Rate (FAR), (ii) False Rejection Rate (FRR), (iii) Equal Error Rate (EER), (iv) Receiver Operating Characteristics (ROC), (v) Success Rate or Hit Ratio, (vi) F-score, (vii) Specificity, (viii) Sensitivity, and (ix) Confusion Matrix (Please provide figures if necessary). 4.5
   (b) Discuss Overfitting vs generalization with necessary example. 2
   (c) What is "cross-validation"? Explain 4 fold cross-validation methods. 2.25

3. (a) Define: (i) input/hidden/output layers, (ii) weights, (iii) activation functions, and (iv) error 2
   (b) Define Hebb rule. Analyze the application of Hebb rule using an autoassociative memory. 3
   (c) Consider the three prototype patterns shown below: 3.75



   i) Use the Hebb rule to design a perceptron network that will recognize these three patterns p_1, p_2 and p_3.
   ii) Find the response of the network to the pattern p_t. Is the response correct?

4. (a) How would you apply a Naive Bayes classifier to characteristics that are categorical? How about features that are numerical? 2.75

(b) Does Naive Bayes fit within the generative or discriminative classifier category? 2

(c) Consider the following data set for Naive Bayesian classification problem. 4

Age	Income	Credit_Rating	Loan (Class)
High	Fair	Low	60,000
Low	Fair	High	70,000
Low	Excellent	High	80,000
High	Fair	High	65,000

Predict the class label for a test sample (A = Low, B = Fair, C = Low) using the Naive Bayesian classification. Note that, you may use the Laplacian correction to avoid computing probability values of zero if you want.

Section B

5. (a) Is it conceivable that during further K-Means rounds, the clustering of observations remains the same? 2

(b) What do you understand by feature space? What is linearly separable and not separable? Explain with example. 2.75

(c) Assume that the objective of the machine learning job is to divide the CGPA of the following students into two clusters. 3.75

A	B	C	D	E	F
3.45	2.78	2.90	3.53	2.25	3.67

We initially designate A and C as the respective centers of each cluster. To display only the final two clusters, use the k-means technique.

6. (a) Write short notes on Decision tree. 2

(b) What is Edit Distance? Write down Edit Distance algorithm. Find out the Edit Distance between 'exercised' and 'executed'. 2.25

(c) What is support vector? Why maximum margin in SVM technique? 2

(d) What is the "kernel trick"? How do you compute M in terms of W and b? 2.5

7. (a) Demonstrate a proof of convergence for the perceptron learning rule, for a single neuron perceptron. 4

(b) Solve the following classification problem with the perceptron rule. Apply each input vector in order, for as many repetitions as it takes to ensure that the problem is solved. Draw a graph of the problem only after you have found a solution. 4.75

$$\left\{p_1=\begin{bmatrix}2\\2\end{bmatrix},t_1=0\right\}\left\{p_2=\begin{bmatrix}1\\-2\end{bmatrix},t_2=1\right\}\left\{p_3=\begin{bmatrix}-2\\2\end{bmatrix},t_3=0\right\}\left\{p_4=\begin{bmatrix}-1\\1\end{bmatrix},t_4=1\right\}$$

Use the initial weights and bias:

$$W(0)=\begin{bmatrix}0 & 0\end{bmatrix} \qquad b(0)=0.$$

8. A produce dealer has a warehouse that stores a variety of fruits and vegetables. When fruit is brought to the warehouse, various types of fruit may be mixed together. The dealer wants a machine that will sort the fruit according to type. There is a conveyer belt on which the fruit is loaded. This conveyer passes through a set of sensors, which measure three properties of the fruit: *shape, texture* and *weight*. These sensors are somewhat primitive. The shape sensor will output a 1 if the fruit is approximately round and a -1 if it is more elliptical. The texture sensor will output a 1 if the surface of the fruit is smooth and a -1 if it is rough. The weight sensor will output a 1 if the fruit is more than one pound and a if it is less than one pound. The three sensor outputs will then be input to a neural network. The purpose of the network is to decide which kind of fruit is on the conveyor, so that the fruit can be directed to the correct storage bin. To make the problem even simpler, let's assume that there are only two kinds of fruit on the conveyor: apples and oranges.
As each fruit passes through the sensors it can be represented by a three dimensional vector. The first element of the vector will represent shape, the second element will represent texture and the third element will represent weight as below.

$$p = \begin{bmatrix} shape \\ texture \\ weight \end{bmatrix}$$

The neural network will receive one three-dimensional input vector for each fruit on the conveyer and must make a decision as to whether the fruit is an *orange* (p_1) or an *apple* (p_2).

(a) Write down the prototype templates for *orange* (p_1) and *apple* (p_2). 1
(b) Design a perceptron to recognize these patterns. 1.5
(c) Design a Hamming network to recognize these patterns. 1.5
(d) Design a Hopfield network to recognize these patterns. 1.5
(e) Test the operation of your networks by applying several different input patterns. Discuss the advantages and disadvantages of each network. 3.25